

IRINGA, Tanzania, United Republic of

WMO: 638870

Lat: 7.669S

Lon: 35.752E

Elev: 1426

StdP: 85.33

Time Zone: 3.00 (E03)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	13.0	4.4	6.1	24.5	5.8	6.8	21.7	12.8	20.8	10.7	19.7	1.6	180	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	11.8	30.2	17.3	29.6	17.5	28.9	17.5	21.0	26.1	20.5	25.5	20.1	25.0	6.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	16.8	22.9	19.0	16.5	22.6	18.6	16.0	22.2	68.1	26.1	66.3	25.5	64.4	25.0	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.9	8.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			8.0	23.4	0.5	1.7	7.6	24.7	7.3	25.7	7.0	26.6	6.6	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	21.9	21.9	21.7	21.2	20.6	19.5	19.1	19.8	20.9	22.3	23.1	22.5	
	DBStd	1.60	1.16	1.01	1.03	0.86	0.86	1.14	1.09	1.10	1.05	1.07	0.94	1.07	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	10	0	0	0	0	0	3	5	2	1	0	0	0	
	CDD10.0	4086	369	333	362	335	327	286	282	303	328	381	392	388	
	CDD18.3	1052	111	100	103	85	69	38	28	46	78	122	142	130	
	CDH23.3	6059	468	497	479	316	278	201	187	278	602	969	1097	687	
	CDH26.7	1029	61	71	59	17	11	11	8	20	89	239	315	129	
Wind	WSAvg	2.9	1.6	1.7	1.4	2.3	3.1	3.6	3.8	3.5	3.6	4.0	3.7	2.1	
Precipitation	PrecAvg	627	133	107	126	67	10	0	0	0	0	4	32	148	
	PrecMax	1089	265	223	283	250	55	11	2	10	7	26	130	361	
	PrecMin	466	21	31	25	10	0	0	0	0	0	0	0	34	
	PrecStd	139	59	47	59	54	13	2	0	2	2	7	31	83	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.6	29.8	29.2	28.3	27.9	28.0	27.8	28.5	30.1	31.1	31.2	30.3	
		MCWB	18.7	18.6	20.0	19.0	17.6	16.9	16.0	16.5	16.4	16.2	17.3	18.2	
	2%	DB	28.1	28.3	28.2	27.1	26.8	26.6	26.2	27.1	28.9	30.1	30.2	29.1	
		MCWB	18.8	19.1	19.5	19.1	17.5	16.3	15.7	16.0	16.1	16.7	17.3	18.2	
	5%	DB	27.1	27.4	27.2	26.1	25.9	25.5	25.2	26.0	27.9	29.1	29.6	28.2	
		MCWB	19.1	19.2	19.7	19.0	17.4	16.0	15.4	15.4	16.0	16.6	17.3	18.2	
	10%	DB	26.1	26.5	26.2	25.2	25.0	24.4	24.2	25.0	26.8	28.1	28.8	27.1	
		MCWB	19.3	19.2	19.5	18.8	17.3	15.7	15.0	15.2	15.7	16.6	17.3	18.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	21.5	21.6	21.0	19.4	18.2	17.6	17.7	17.8	19.1	20.7	20.8
			MCDB	25.9	26.5	27.3	25.4	24.6	24.8	25.3	26.1	27.3	27.4	27.5	26.0
2%		WB	20.5	20.7	20.8	20.2	18.5	17.4	16.6	16.7	17.2	18.2	19.6	20.1	
		MCDB	25.2	25.6	26.2	24.9	24.5	24.7	25.2	25.6	26.6	26.9	26.1	24.8	
5%		WB	20.0	20.2	20.3	19.6	18.1	16.7	16.0	16.1	16.7	17.7	18.9	19.6	
		MCDB	24.6	25.2	25.5	24.4	24.4	24.1	24.3	24.8	26.0	26.5	25.3	24.3	
10%		WB	19.6	19.7	19.7	19.2	17.6	16.1	15.3	15.6	16.2	17.2	18.4	19.2	
		MCDB	24.0	24.5	24.6	24.0	23.8	23.4	23.2	24.0	25.5	26.1	25.2	24.0	
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.7	9.7	9.2	9.9	11.1	11.4	11.5	12.5	12.3	11.8	9.9	
		MCDBR	10.8	11.5	11.3	10.5	11.5	12.4	13.0	13.0	14.1	13.9	13.5	11.5	
		MCWBR	3.9	4.6	4.7	4.4	4.8	5.0	5.4	4.9	4.6	4.4	4.2	3.5	
	5% WB	MCDBR	9.0	9.8	10.1	9.2	10.0	11.8	12.7	12.1	12.9	12.5	11.3	9.3	
		MCWBR	3.9	4.4	4.7	4.4	4.6	5.2	5.6	5.0	4.8	4.5	4.2	3.5	
Clear-Sky Solar Irradiance	taub	0.385	0.380	0.353	0.314	0.313	0.310	0.326	0.357	0.362	0.371	0.349	0.361		
	taud	2.435	2.458	2.551	2.753	2.646	2.570	2.469	2.358	2.351	2.336	2.451	2.490		
	Ebn at Noon	954	955	966	978	952	939	929	922	943	952	982	974		
	Edh at Noon	122	119	107	84	89	93	105	122	129	133	119	115		
All-Sky Solar Radiation	RadAvg	5.52	5.77	5.66	5.28	5.09	5.25	5.41	5.80	6.49	6.64	6.38	5.76		
	RadStd	0.42	0.30	0.28	0.26	0.28	0.17	0.25	0.24	0.26	0.25	0.37	0.40		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page